Development of Soil Resource Mapping Device for Extreme Environments

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ABSTRACT

Measurements within permanently shadowed craters and along steep embankments on extraterrestrial surfaces pose extreme challenges since astronauts and rovers cannot access them. However, these areas offer the potential for the greatest scientific returns. Researchers at the NASA Langley Research Center (LaRC) invented a deployable wireless micro-spectrometer that can access those challenging areas and telemetry data to a receiver safely outside those extreme environments. The micro-spectrometers are designed for deployment by an astronaut, a rover, or a lander flying overhead. This device can also be installed on rover tires, under the astronaut's shoes, or on a cane stick. The bullet-like, consumable, micro-spectrometer can penetrate into soil to spectrally identify the components of soil, such as water, He-3, or other minerals. The signals from the soil assay data are transmitted to a mother station through a telemetry system. This LaRC developed micro-spectrometer bullet consists of micro-spectrometer optics with an all-imbedded, burst-mode, light-emitting-diode ultraviolet (LED UV) light source, a super-capacitor with control electronics, and telemetry electronics. Prototypes have been fabricated to demonstrate spectral assay of soil components. Further, maturation of this technology would be necessary for demonstrations on the Moon.

INTRODUCTION

Micro-spectrometer (MSM) is based on the Fresnel diffraction principle which allows high resolution imaging of spectral signal within a very short distance of light passage through micro-differential ring grating or micro-differential linear grating. The concept and studies of MSM initially invented at NASA LaRC are illustrated by Park (US Patent 2010), Park (2005), Park (2008), Park (2013), and 6 U.S. patents (2008, 2011, 2012) and also fabricated as a prototype to test its functional performance against the luminescence or fluorescence of bio-component such as dopamine. Figure 1 shows the MSM prototype used for the measurement of spectral signal from dopamine. The test data shows the functioning performance of the MSM in measuring the molar concentration of dopamine as shown in Figure 2. When the molar density of dopamine increases from nothing to 0.1μM, 1μM, and 10μM, the signal outputs are inversely decreased due to the absorption decay of the original signal. For the soil assay application, the MSM, along with other parts, is packaged into a bullet shape which can be delivered to a long distance by shooting.

The assay of lunar and Martian soils by actual measurements on plain fields, within permanently shadowed craters, and along steep embankments will help create a mineral map which is very essential for the successful mining and excavation of resources under NASA In-Situ Resources Utilization (ISRU) program. However, spectral mapping of permanently

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shadowed craters, and along steep embankments pose extreme challenges since astronauts and rovers cannot readily access them.

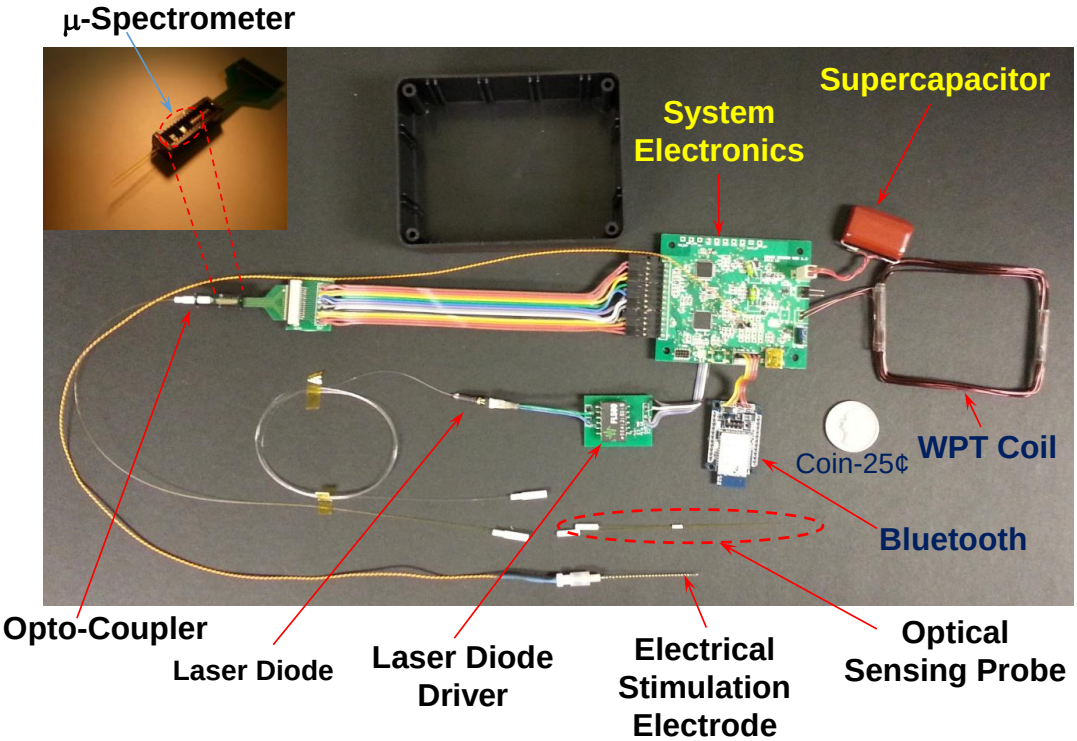

Figure 1. A prototype micro-spectrometer (MSM).

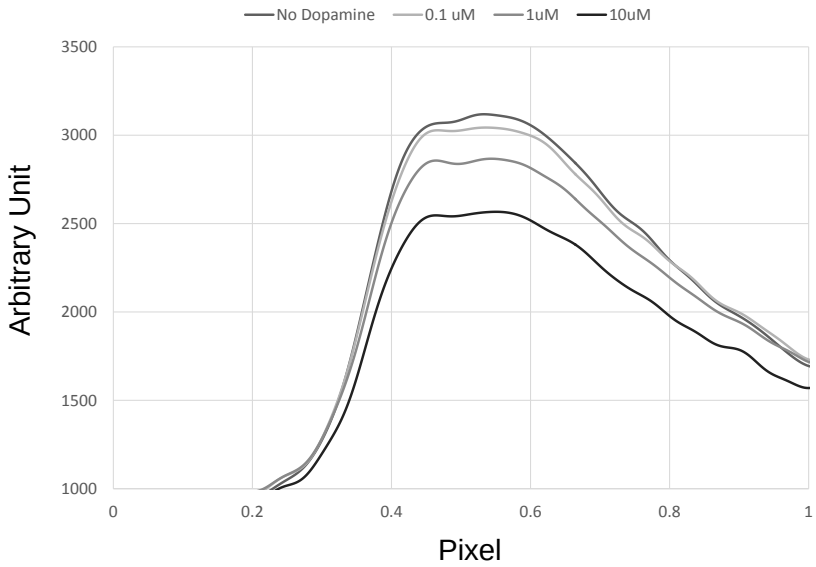

Figure 2. Test result by a prototype micro-spectrometer (MSM) against dopamine molar concentration

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To address this shortcoming, a new device concept offers a shootable micro-spectrometer bullet for direct access to the areas with geologically ragged formations and extreme environments. This new concept device transmits measured signals wirelessly back to the main station safely outside those extreme environments through an imbedded telemetry system. Such a scenario is shown in Figure 3. Using multiple devices simultaneously allows large areas to be interrogated quickly and measurements to be coordinated within areas of interest.

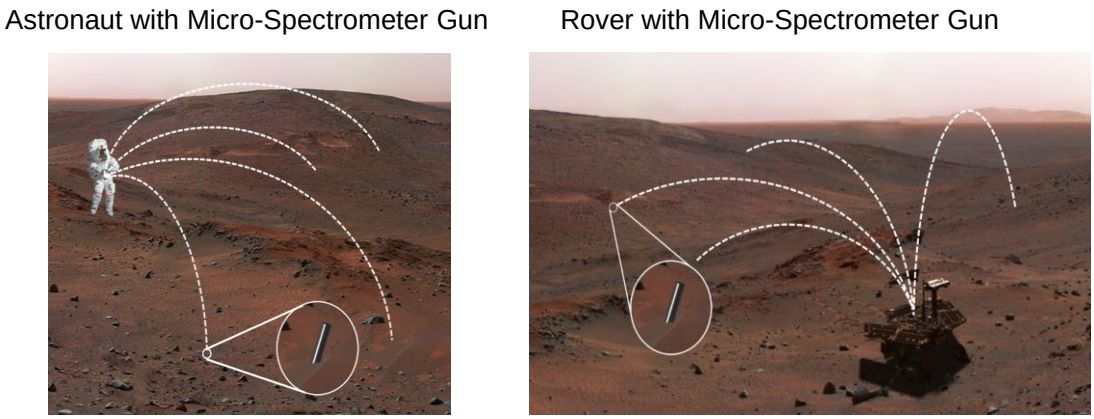

Figure 3. Bullet-like consumable micro-spectrometers are shot out broadly to assay organic and inorganic components of Lunar/Mars/Asteroids soils.

DESCRIPTION OF BULLET MICRO-SPECTROMETER

The micro-spectrometers are designed for deployment by an astronaut, or a rover, or a lander flying overhead. Many of these devices can be shot by astronauts, or by rovers, or even from satellites for wide area coverage of soil assay, as shown in Figure 3. This device can also be installed on rover tires, under the astronaut's shoes, or at the bottom of a cane stick or on the tumbleweed rover (see Figure 4).

The bullet-like consumable micro-spectrometer can be shot as a projectile to a target area and penetrate to a certain depth in the soil and, at the same time, be triggered to spectrally identify the components of soil, such as water, He-3, or other minerals, as it penetrates. The spectral data collected from the soil assay are transmitted to a main station through a telemetry system. This LaRC developed micro-spectrometer bullet by Choi (NASA Invention 2019) consists of micro-spectrometer optics with an all-imbedded burst-mode LED UV light source, a super-capacitor with control electronics, and telemetry electronics, as shown in Figure 5.

The unit cell of the micro-spectrometer is represented by the block indicating 'Optics/Sensing unit' in Figure 5 and is distinguished by two different modes based on the input signal injection pattern: (a) parallel injection and (b) vertical injection, shown as (a) and (b), respectively, in Figure 6. Regardless of its injection orientation, the spectral signal from target material, after impinging on and passing through the differential linear Fresnel grating, either (c) or (d) in Figure 6, will shine onto the pixel array of the imaging sensor. In this process, the photons of the spectral signal of fluorescence or luminescence from target material are diffracted through Fresnel grating and then fall onto a specific pixel depending on the wavelength of spectral signal since the diffraction through the Fresnel grating is wavelength (or photon energy) dependent. To differentiate the spectral lines further after diffraction, a high pixel density of the

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image sensor is required to receive the specified wavelength of the incoming signal as shown in Figure 7. Thus, a high resolution Fresnel grating requires a high pixel density sensor to separate spectral lines to identify different soil components. The spectral data obtained by addressable pixel element is digitized through the analog to digital converter (ADC) and micro-controller. Then the telemetry unit transmits the digitized data to a receiving station where the data are collected and processed for image construction of a mineral map or to show the distribution of other elements.

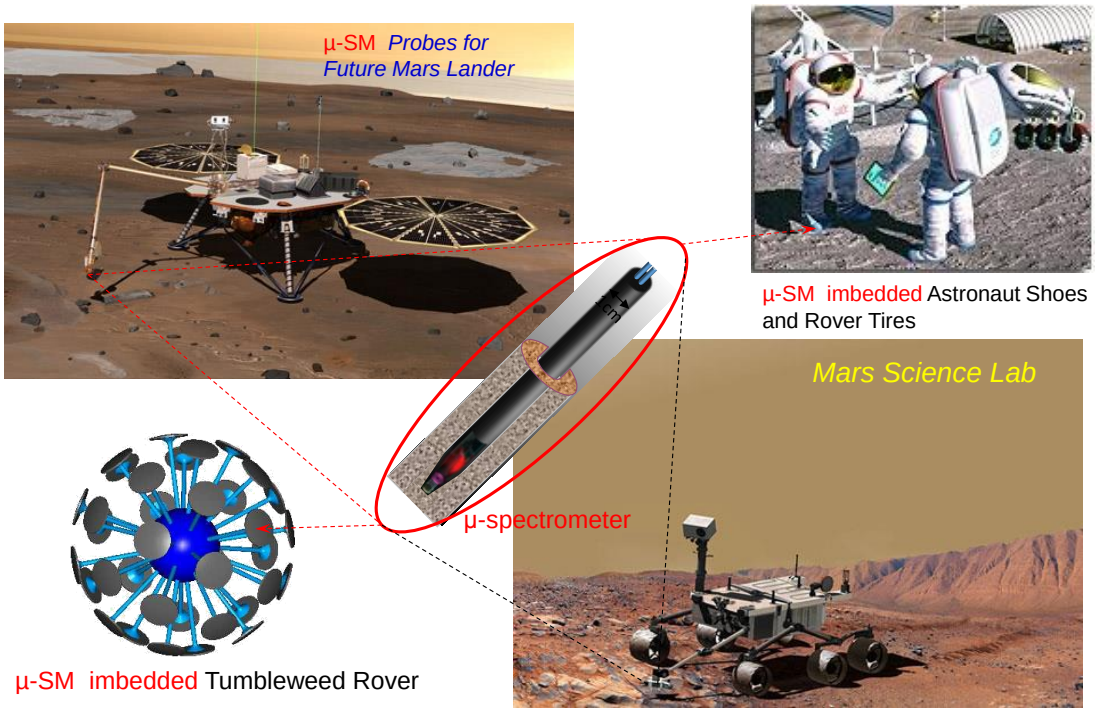

Figure 4. Bullet micro-spectrometers can be attachable to many different places to effectively assay organic and inorganic components of Lunar/Mars/Asteroids soils.

Upon shooting, a bullet micro-spectrometer flies along a trajectory path to reach the target surface and penetrates into the regolith soil by its momentum. At the moment of penetration, a burst-mode LED or a laser diode is automatically turned on by an impact switch to generate an intensive deep UV (DUV) or vacuum UV (VUV) light. The burst-mode LED or a laser diode is powered by the super-capacitor that stores energy. The DUV or VUV from a burst-mode LED or a laser diode in a pulse mode illuminates and excites the soil components to create a singlet or triplet transition according to the amount of absorbed photon energy. Since the pattern of optical absorption and associated quantum transition that emanates fluorescent or luminescent emission varies from the element or molecular structure of the material, these transitions are often used to identify the spectral signature of materials. Figure 8 shows a micro-spectrometer bullet that is penetrated into the soil. As explained above, the fluorescence or luminescence of emission spectra from the excited soil components generates the spectral signal picked up by a sensor array which is subsequently digitized, and transmitted wirelessly through an onboard telemetry system and antenna at a receiving station. The data logger and analyzer at the receiving station analyze the transmitted data from the bullet-like micro-spectrometer to illustrate spectrally the resolved signature of the chemicals detected in the soil.

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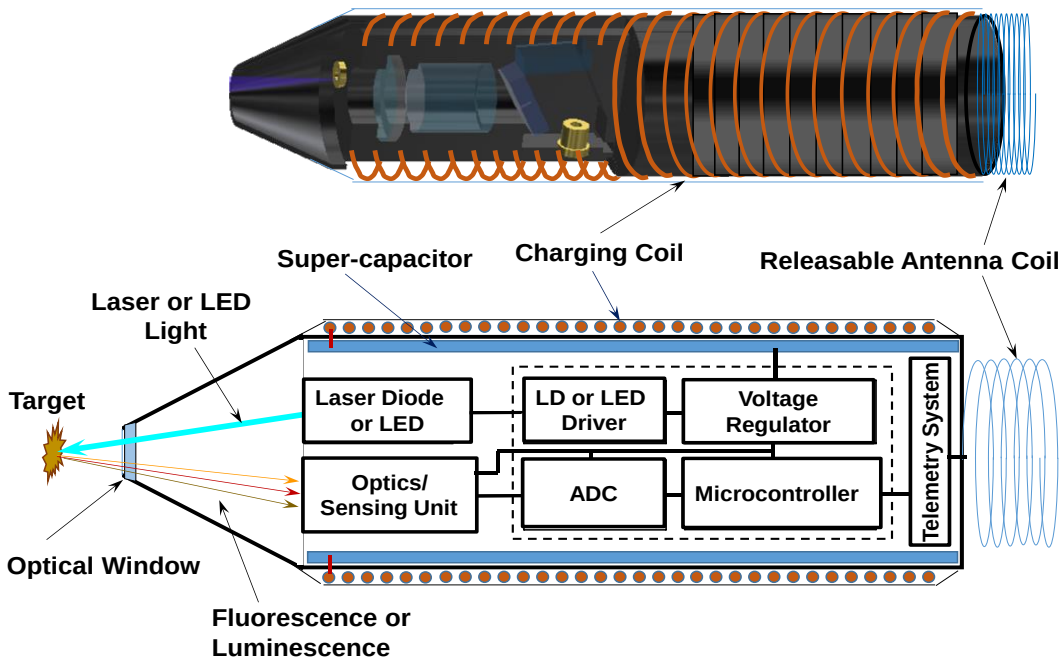

Figure 5. Block diagram of bullet micro-spectrometers.

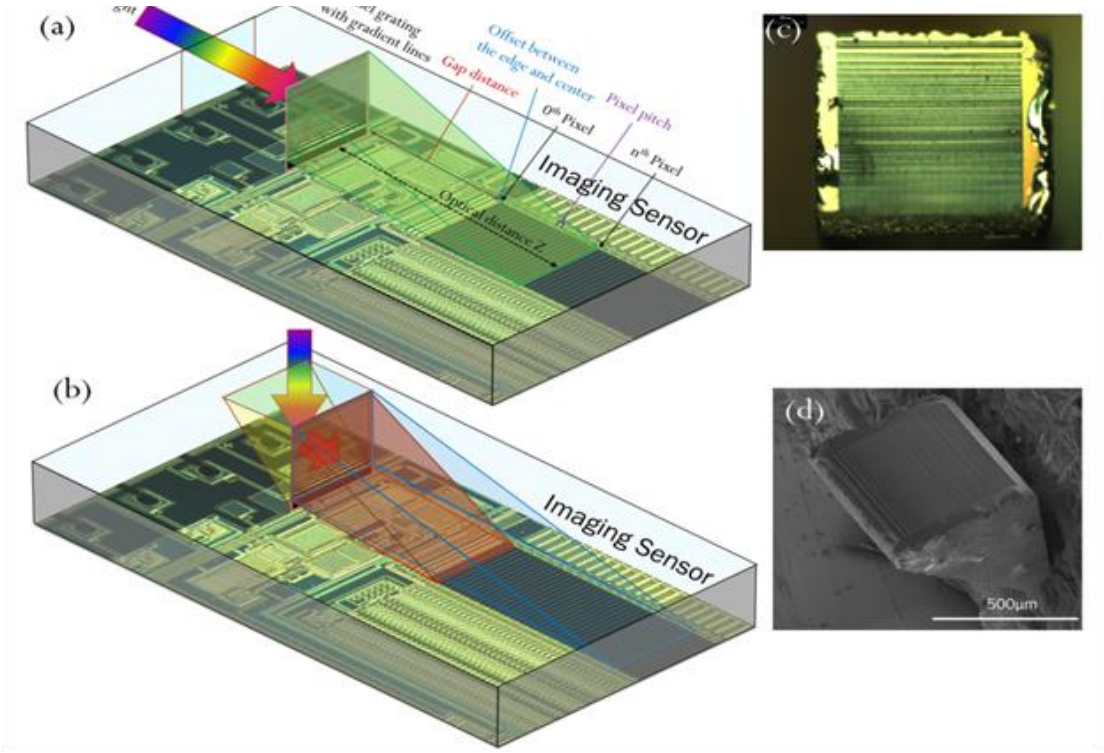

Figure 6. Micro-spectrometer chip models: Parallel injection mode cell (a) with linear differential Fresnel grating (c) and Vertical injection mode cell (b) with right angle prism linear differential Fresnel grating (d).

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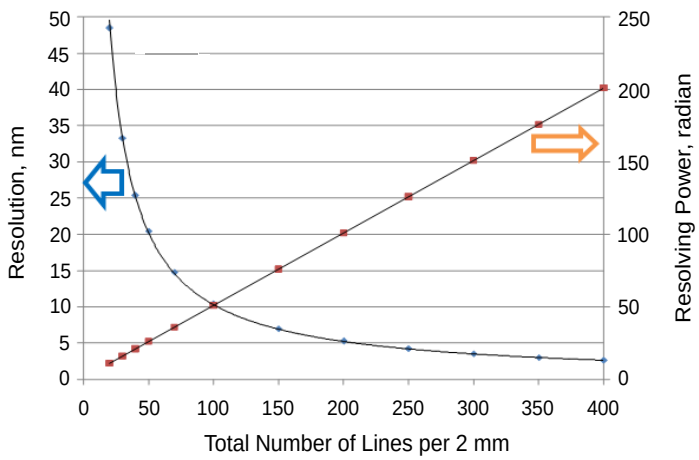

Figure 7. Resolution and resolving power of differential Fresnel grating based on total number of lines per 2 mm height.

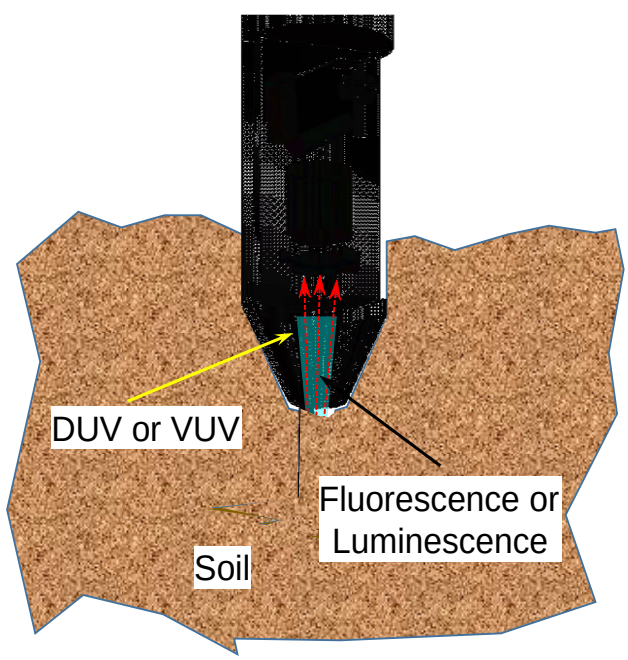

Figure 8. DUV or VUV from a burst mode of LED or from a diode laser illuminates target soil to generate fluorescence or luminescence emission.

Figure 9 shows a compressed gas dispenser. When a bullet MSM from the magazine is loaded into a shooting position, the compressed gas is released by the trigger, opening the valve. When the valve is opened, the compressed gas expands and pushes a readied bullet MSM through the barrel. The bullet MSM passes through the array of magnets positioned circumferentially on the barrel, shown in Figure 9, where a charge coil generates an electric current to charge the super-capacitor and accelerate the bullet MSM out of the gun barrel.

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When the bullet MSM leaves the end of barrel, the antenna coil is stretched and straightened by its own stretching force from the coiled spring. In this way, as the bullet MSM penetrates powdered soft soil, the stretched antenna coil can still appear above the soil and communicate with a receiving station as shown in Figure 10.

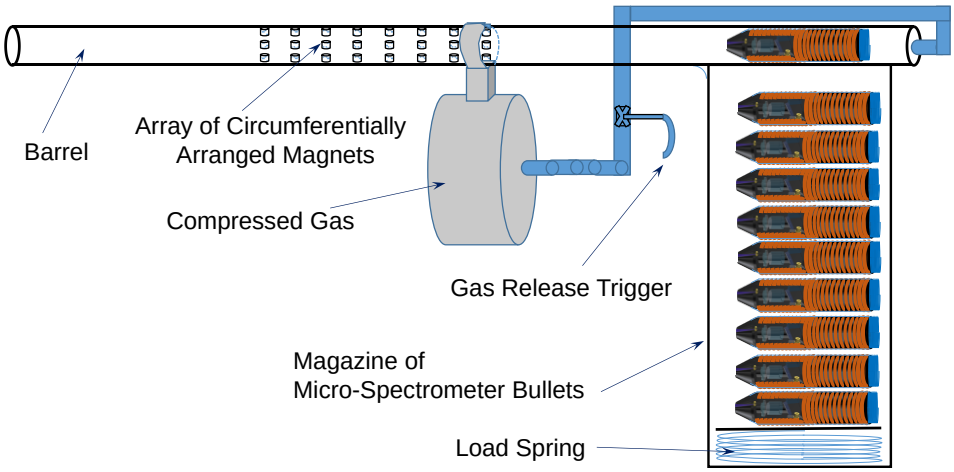

Figure 9. Conceptual description of a compressed gas gun to shoot bullet micro-spectrometers.

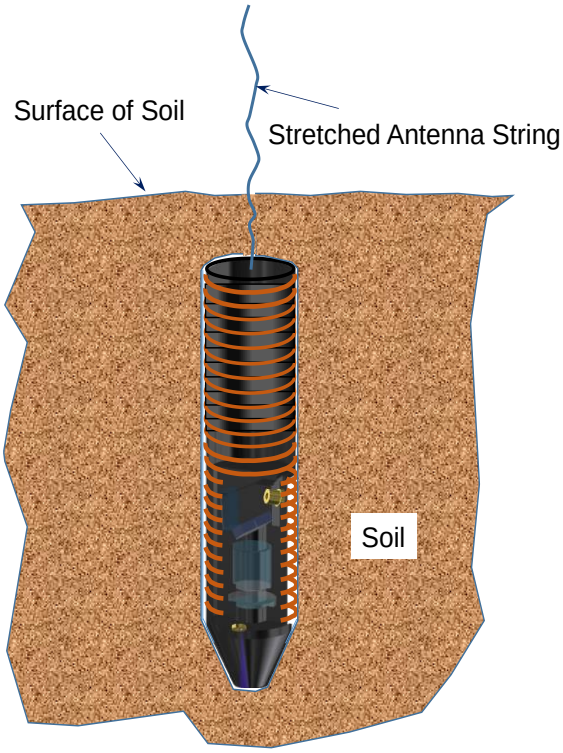

Figure 10. Deployed a string antenna when MSB pokes deep into soil.

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Other shooting methods include a mortar-shell which has gun-powder at the tail of bullet MSM. In this case when a bullet MSM is loaded, the gun-powder is ignited to propel the bullet MSM through magnetic field of gun barrel. Otherwise, the rest of the device concept and process is the same as that of compressed gas propelled bullet MSM described above.

Another deployment method includes a pencil rocket connected at the tail of bullet MSM. In this case when a bullet MSM is loaded, a pencil rocket is ignited to accelerate the bullet MSM through the magnetic field within barrel. Otherwise, the rest of the device concept and process is the same as that of compressed gas propelled bullet MSM.

RESULTS

We have developed the micro-spectrometers as a neural probe application (see Figure 1). There are several critical elements that need to be refined for a deployable class of bullet micro-spectrometer (MSM). These are the avalanche photodiode (APD) pixel array to enhance the sensitivity of photo-sensor of micro-spectrometer bullet, the super-capacitor to supply sufficient power for APD and telemetry system, the gun barrel or booster rocket with electric charger for super-capacitor, and the telemetry system for data transmission.

CONCLUSIONS

The refinement of micro-spectrometer systems illustrated above will raise the technology readiness level (TRL) 5~7. Numerous soil assay experiments will be compared with the spectral database of known soil components compiled by U.S. Geological Survey (USGS) for validation of the device. Once the validation process is over, the bullet micro-spectrometer with gun or launching device will be prepared for deployment packaging as a final process.

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